

Team Journey

AquaSensePhase1—AutomaticTouchlessWaterDispenser

Team Details

Role	Name
Team Member	Kushwanth
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Mentor / Guide	Mahesh Devagiri

Chronological Team Journey

1. Idea

The team identified the need for a hygienic, convenient, and controlled method of dispensing water without requiring direct hand contact.



2. Initial concept

The initial concept combined hand detection, an Arduino controller, a display, servo motors, and a mechanical valve or tap.



3. Circuit design

The team planned the connections between the Arduino Uno, HC-SR04 ultrasonic sensor, I2C LCD, servo motors, and power supply.



4. Hardware selection

Components were selected according to availability, ease of programming, operating voltage, mechanical requirements, and suitability for a Phase 1 prototype.



5. Arduino implementation

The Arduino program was developed to initialize the hardware, measure distance, compare it with the detection threshold, control the servos, and display system status.



6. Sensor + LCD + servo integration

The ultrasonic sensor, LCD, and two SG90 servos were integrated and tested as one control system. The system was configured to detect a hand within 15 cm and dispense water for 3 seconds.



7. 3D modelling

A 3D model was developed to study the enclosure, component placement, valve mechanism, and overall appearance of the dispenser.



8. Two-box concept

An early enclosure concept used two separate boxes to organize the electronics and the dispensing or mechanical section.



9. Design evaluation

The two-box design was reviewed for size, wiring, usability, assembly effort, maintenance, stability, and overall appearance.



10. Single-box redesign

Based on the evaluation, the design was refined into a single-box concept to achieve a more compact, integrated, and user-friendly prototype.



11. Website development

A project website was developed to present the system concept, features, working principle, components, and project progress.



12. Documentation

The team prepared the source code, pin configuration, component specifications, system working principle, and project records.



13. Phase 1 prototype

The Phase 1 prototype brought together the electronics, control software, mechanical actuation, enclosure concept, and supporting documentation for demonstration and future improvement.

Journey Flow Summary

Stage	Outcome
Idea → Initial concept	Defined the touchless dispensing objective and proposed system architecture.
Circuit design → Hardware selection	Mapped the electrical interfaces and selected practical prototype components.
Arduino implementation → Integration	Created the control logic and combined sensing, display, and valve actuation.
3D modelling → Design evaluation	Explored enclosure alternatives and assessed the two-box concept.
Single-box redesign → Website development	Improved physical integration and communicated the project digitally.
Documentation → Phase 1 prototype	Recorded the work and prepared the integrated prototype for demonstration.

Work Session Photos

No work-session photos were available in the provided materials for insertion into this document. Photos can be added later under this section, such as circuit assembly, Arduino testing, 3D modelling, enclosure fabrication, and prototype demonstrations.

Photo placeholder	Suggested caption
Photo 1	Circuit design or wiring work session
Photo 2	Arduino, sensor, LCD, and servo integration
Photo 3	3D modelling or enclosure design
Photo 4	Phase 1 prototype demonstration

Team Reflection

The project progressed through repeated cycles of planning, implementation, testing, evaluation, and redesign. The transition from the two-box concept to the single-box design demonstrates how practical feedback and design evaluation guided the development of a more integrated Phase 1 prototype.